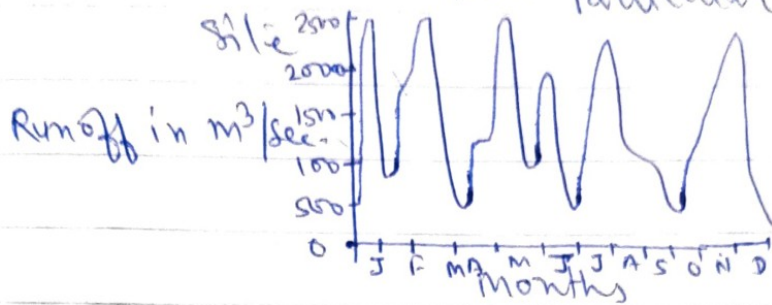
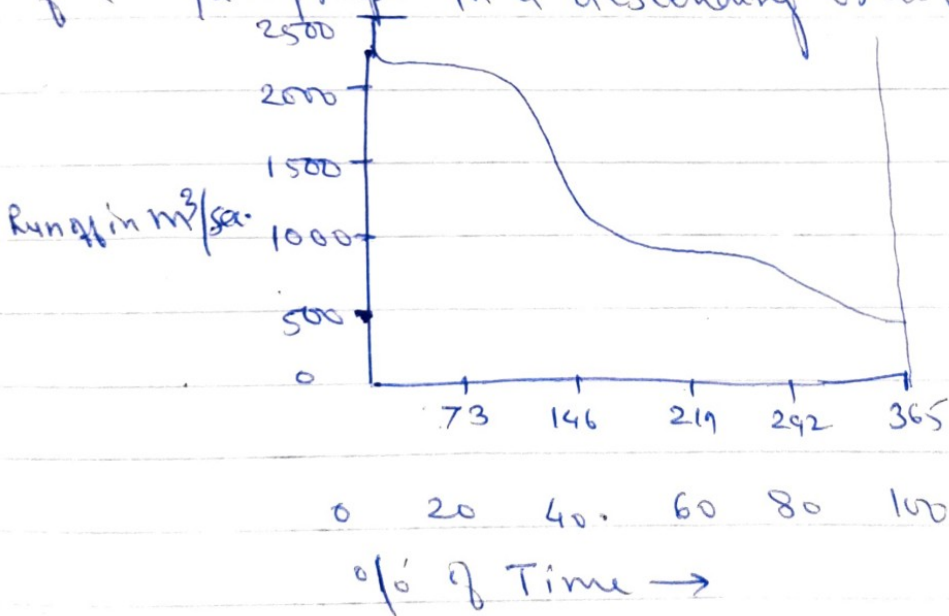


Hydroelectric power plants

Hydrograph → Variation of stream flow in $\text{m}^3/\text{sec.}$ with time for a Particular River



Flow duration curve → Rearrangement of all the stream flow elements of a hydrograph in a descending order



Each point on a flow duration curve shows the percentage time during the period when the flow was equal to or greater than the given flow.

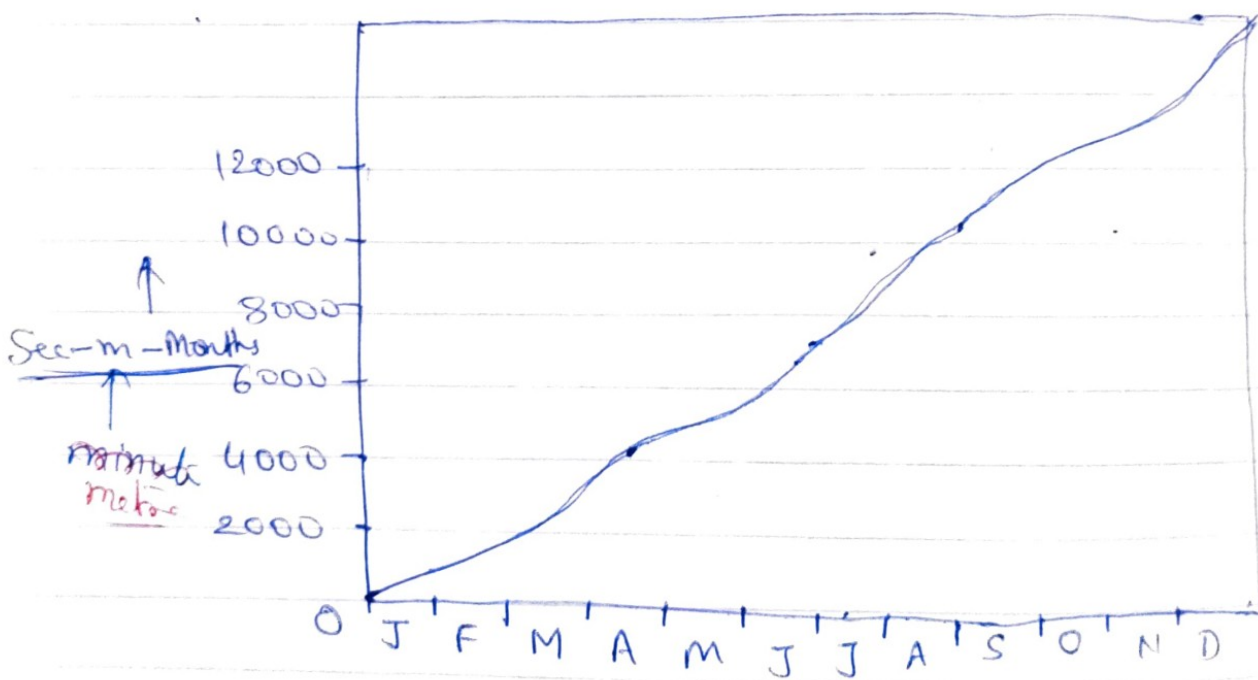
Mass curve

Mass curve indicates the Total Volume of runoff in cubic meters up to a certain time.

→ A mass curve is obtained from the records of average monthly flows.

→ The slope of the curves at any point shows the rate of flow at that ~~time~~ time.

→ If Rainfall is uniform throughout the year, the mass curve could be a straight line having uniform slope.



Mass curve are used in estimating the capacity of storage reservoir in hydroelectric projects.

ordinate → may be in Sec-m-day
Sec-m

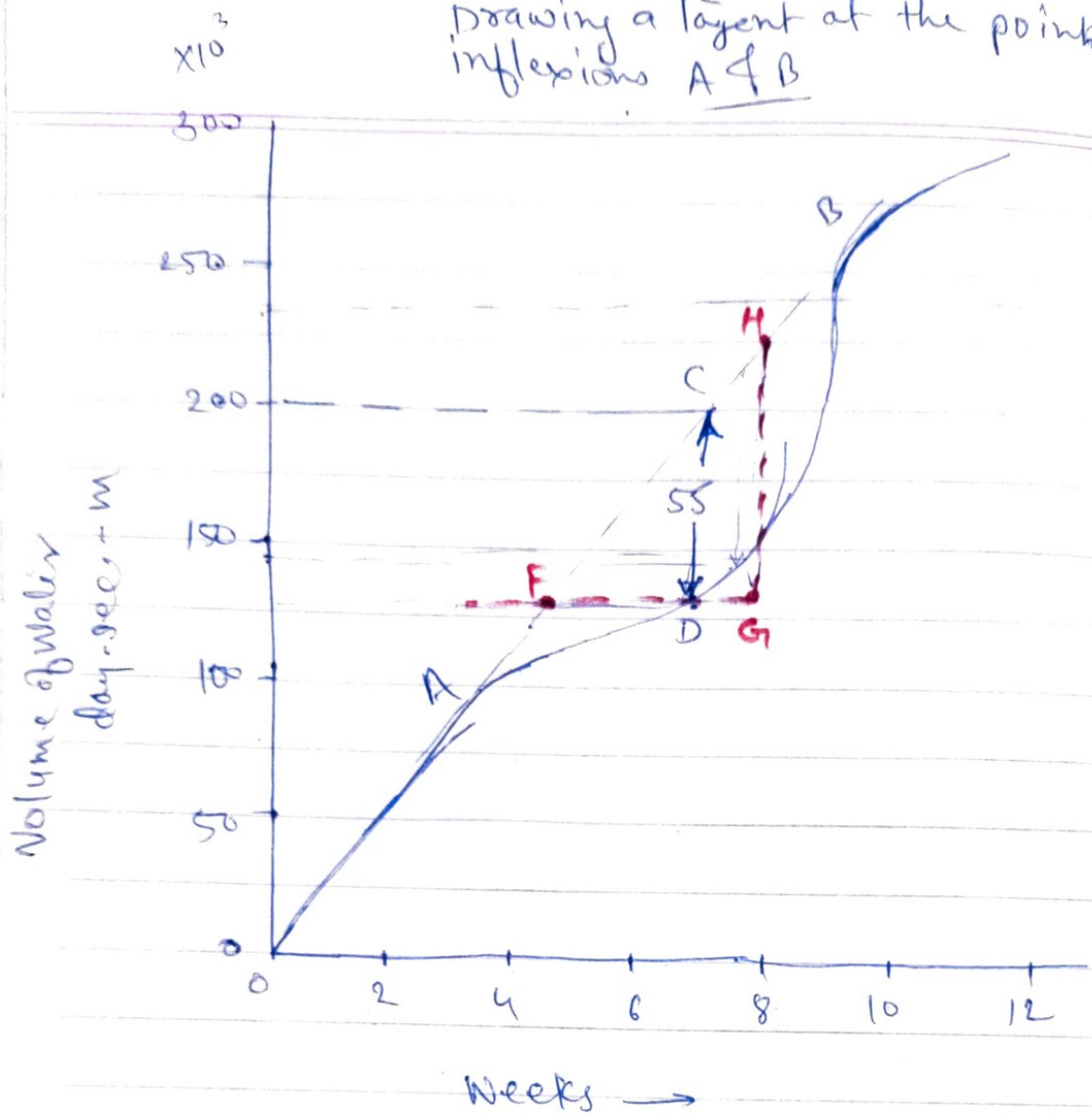
Second-m-day $\rightarrow$ means that
 $\rightarrow$ Flow collected at the rate of
1 cubic metre per second
_{m³}
 for one day or one month.

$$1 \text{ second-metre-day} = 1 \times 24 \times 60 \times 60 = 86400 \text{ Cubic meters}$$

How to Obtain mass curve:- (For this cumulative
 Volume of Water that can be stored Week after Week
 must be found.

week	(Col-2) Weekly Flow (m ³ /sec.)	(Col-2 x 7) Weekly Volume day second metres _{m³}	summation
1	6000	42000	42000 42000
2	5000	35000	77000
3	4000	28000	105,000
4	2000	14000	119,000
5	1000	7000	126,000
6	500	3500	129,500
7	1000	7000	136,500
8	4000	28000	164000
9	8000	56000	220500
10	5000	35000	255,500
11	2000	14000	269,500
12	1000	7000	276,500 276,500

Drawing a tangent at the points of inflexions A & B



$$\text{Capacity of Reservoir} = 55 \times 10^3 \text{ day-second-meter}$$

$$\frac{HG}{FG} = \frac{87.5 \times 10^3}{4 \times 7} = 3120 \text{ m}^3/\text{sec.}$$

CD $\rightarrow$ max. ordinate between the slope line and the mass curve, gives the capacity of Reservoir.

when ~~the~~ a reservoir of this capacity has been built, it is possible to have a flow of ~~this capacity~~ from the river at the rate of $3120 \text{ m}^3/\text{sec.}$

If the rate at which water is drawn is higher than this, the reservoir level goes down.

→ To fill fill the seasonal variation in stream flow.

Storage → It indicates the building up of a low natural flow of the stream to a uniform discharge.

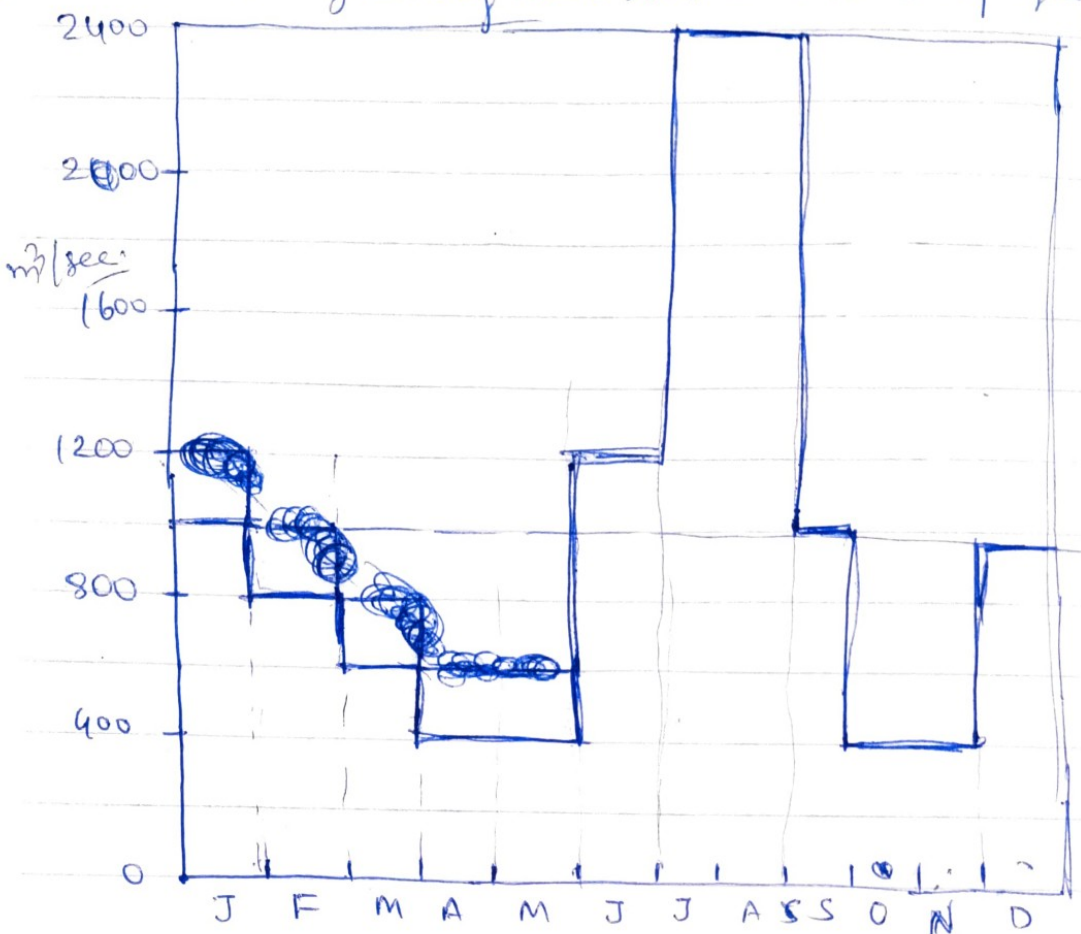
Pondage → Indicates the regulation of the resulting uniform flow, or natural flow if there is no storage.

→ The power demand fluctuates with time, If the power plant is away from the storage reservoir, a small pond is needed near the power plant to meet the hourly changes in demand. (load)

Example — The average rate of inflows during 12 months for a river are as follows:
Under:

Jan.	1000 cu-m/sec.	July	2400 cu-m/sec.
Feb.	800 "	Aug.	1000 2400 cu-m/sec.
March.	600 "	Sept.	1000 "
April	400 "	Oct.	400 "
May	400 "	Nov.	400 "
June	1200 "	Dec.	1000 "

Plot the hydrograph. Find the average inflow and the power that can be developed at an effective head of 150m and a turbine-gen. efficiency of 0.85. Find the capacity of storage plot hydrograph reservoir based on above one year data neglecting the losses due to seepage evaporation etc.



Average rate of flow

$$\begin{aligned}
 &= \frac{1}{12} (1000 + 800 + 600 + 400 + 1200 + 2400 + 2400 + 1000 + 400 + 400 + 1000) \\
 &= 1000 \text{ cum/sec.}
 \end{aligned}$$

$$P = \frac{735.5}{75} \times 1000 \times 150 \times 0.85$$

$$= 1250.35 \text{ MW.}$$

Since average flow is $1000 \text{ m}^3/\text{sec}$. This is also the demand line

The storage capacity of reservoir is evidently the excess flow during June, July and Aug. i.e. Total of 3000 ccm-month

$$\text{or } 30 \times 24 \times 60 \times 60 \times 3000 = 7.776 \times 10^9 \text{ m}^3$$

Site selection of Hydro electric plants

① Availability of water → The river runoff data pertaining to many years

should be available so that estimate of the power potential of the project can be made.

data → average flows, min^m and max. flows and their periods

② Water storage. → Because of wide fluctuations in stream flows, storage is needed in most ~~of~~ hydro-projects to store water during high flow periods and use it during lean flow periods.

Storage arrangement require topographical and geographical investigations. It may be located near the power plant or some distance away from it.

③ Head of water → An increase in effective head reduces the quantity of water to be stored and handled by penstocks, turbines etc. and therefore the capital cost of the plant is reduced.

The head of water depends on topography of the area.

④ Geological investigations → Needed to see the foundation rock for the dam and other structures is firm, stable, impervious and strong enough to withstand water thrust and other stresses.

The area should also be free from earthquakes.

⑤ Water pollution → Polluted water may cause excessive corrosion and damage to metallic structures. So it's necessary to see the water is of good quality.

⑥ Sedimentation → The capacity of storage reservoir is reduced due to the gradual deposition of silt. Silt may also cause damage to turbines blades.

~~Hydro project~~

⑦ Environmental effects → Hydro projects
Submerge huge areas and many villages. As
such the environmental effects are also important.
The site should ensure safe and pleasing
surroundings

⑧ Access to site : — Construction of a hydro
projects involves transport of huge
amount of cement, steel, other building material
and heavy machinery. The site selected should be
such that the railway line and roads can be
constructed and material and machinery transported.

— x —

Power duration curve : — If the head at which the
discharge of water is available is known, the
possible power that can be developed from the
water can be determined. Thus the flow duration
curve can be converted to a power duration
curve with different scales

$$P = \frac{0.736}{75} W Q H \eta, \text{ (Kwatts)}$$

W = density of water 1000 kg/m^3

Q = discharge rate m^3/sec

H = Head in m.

η = efficiency of Turbine and generator.

Classification of Hydro plants according to water flow Regulation

① Run off river plants without Pondage

→ These plants are located such that they use water as it comes (directly), without any pondage or storage. There is no control on flow of water. During periods of high flows or low loads, water is wasted. During lean flow periods, the plant capacity is very low.

In such sites the water is mainly used for irrigation ~~and~~ or navigation and power plant is only incidental. Such plants can be built at a considerably low cost but the head available and amount of power generated are generally low.

about 72 MW

② Run off river plant with pondage →

Pondage Refers to storage at ^{the} plant to take care of hour to hour fluctuations in load on the station. Pondage increases the firm capacity of the station provided the floods do not raise the tail race water level thus reducing the effective water head and plant output.

When plenty of water is available, these plants can be used as base load plants.

When stream flow decreases, these plants can be made to work as peak load plants.

252 MW

- ← Most of the plant in world belongs to this category
- ③ Reservoir Plants → When water is stored in a big reservoir behind a dam, it is possible to control the flow of water and use it most effectively. Storage increases the firm capacity of the plant.
- Example → Bhakra plant in India.

Classification according to Load

- ① Base load Plants → They feed the base load of the system. Thus they supply almost constant load throughout and operate on a high load factor. Base load plants are usually of large capacity. Runoff river plants ~~without~~ pondage and reservoir plants are used as base load plants.
- ② Peak load plants → They are meant to supply the peak load of the system. Runoff river plants with pondage can be used as peak load plants during lean flow periods. Reservoir plants can, of course be used as peak load plants also. Peak load plants have large seasonal storage. They store water during off peak periods and are run during peak load periods. They operate at a low load factor.
- A special type of peak load plant is pumped storage plant.

Hydro plants Classification

Classification according to capacity of plants

- (i) Micro hydel plants $\rightarrow$ up to 100 kW
- (ii) Mini hydel plants $\rightarrow$ up to 1 MW.
- (iii) Small " " $\rightarrow$ up to 10-15 MW
- (iv) Medium " " $\rightarrow$ 15 to 100 MW.
- (v) High capacity " $\rightarrow$ 101 to 1000 MW.
- (vi) Super hydel " $\rightarrow$ above 1000 MW.

Mini and Micro plants: used for supplying ^{isolated} pockets of Villages where grid is not accessible

They are attractive, as the energy generated can be consumed near the source itself, without a big network of subtransmission systems.

Small hydro plants (SHP) are being developed, which is an attractive renewable source of energy in remote and hilly regions isolated from main grids.

- Advantages of SHP:
- (i) Readily accessible source of renewable energy.
 - (ii) Can be installed making use of water head as low as 2 metre and above
 - (iii) Does not involve setting up of large dams
 - (iv) Least polluting
 - (v) Limited initial investments and short gestation periods.
 - (vi) Reduced transmission losses.

discussed in section 8.34.

8.14. CLASSIFICATION ACCORDING TO HEAD

(1) *Low head plants.* When water head is less than 30 m, the plant is called a low head plant. A dam or barrage across the river creates the necessary head. The power plant is located near the dam and, therefore, no surge tank is needed. Either one half of the barrage has regulating gates for discharge of surplus water while the plant is in front of the second half or the plant is constructed by the side of the river. Francis or Kaplan turbines are used.

(2) *Medium head plants.* Medium head plants operate at heads between 30 and 100 metres. An open channel brings water from main reservoir to the fore-bay from where penstocks carry water to the turbines. Francis or Kaplan turbines are used.

(3) *High head plants.* The plants operating at heads above 100 m are generally classified as high head plants. The civil works for these plants include dam, reservoir, tunnel, surge tank and penstock. Generally Francis turbines are used for heads below 200 m and Pelton turbines for still higher heads.